TELCO CUSTOMER CHURN ANALYSIS

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1. Executive Summary & Business Context

Customer retention is a critical profitability driver for subscription-based telecommunications companies. Customer churn directly reduces recurring revenue while increasing acquisition and operational costs.

This engagement was conducted to analyse customer behaviour, identify churn drivers, and develop actionable strategies for improving retention performance.

Business Problem

Management lacks clear visibility into:

- Which customers are most likely to churn
- Key behavioural and financial drivers of attrition
- How retention initiatives should be prioritized

Project Objective

Develop a data-driven framework to:

- Understand churn behaviour
- Identify high-risk customer segments
- Predict churn likelihood
- Enable proactive retention decision-making

Headline Findings

- Overall churn rate: 26.58%
- Nearly 1 in 4 customers leave the service
- Month-to-month contracts drive highest churn risk
- Higher monthly pricing correlates with churn
- Customer loss occurs primarily in early lifecycle stages

Business Outcome

The analysis provides management with a structured analytical foundation to transition from reactive customer loss toward predictive retention management.

2. Dataset Overview

The Telco dataset contains customer demographic, service usage, billing, and contractual information.

Metric	Value
Total Customers	7043
Variables	21
Target Variable	Churn

Data Categories

- Demographics
- Services subscribed
- Contract details
- Billing & revenue data

```python
# Loading Dataset

df = pd.read_csv("../01 Raw Data/Telco-Customer-Churn.csv")
df.head(3)
```

Python

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtection	TechSupport	StreamingTV	StreamingMovies
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	No	No	No	No
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Yes	No	No	No
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	No	No	No	No

3 rows × 21 columns

```python
# Understand Dataset Structure

df.info()
```

✓ 0.0s

```
<class 'pandas.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   str
 1   gender            7043 non-null   str
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   str
 4   Dependents        7043 non-null   str
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   str
 7   MultipleLines     7043 non-null   str
 8   InternetService   7043 non-null   str
 9   OnlineSecurity    7043 non-null   str
 10  OnlineBackup      7043 non-null   str
 11  DeviceProtection  7043 non-null   str
 12  TechSupport       7043 non-null   str
 13  StreamingTV       7043 non-null   str
 14  StreamingMovies   7043 non-null   str
 15  Contract          7043 non-null   str
 16  PaperlessBilling  7043 non-null   str
 17  PaymentMethod     7043 non-null   str
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   str
 20  Churn             7043 non-null   str
dtypes: float64(1), int64(2), str(18)
memory usage: 1.1 MB
```

```python
# Check Missing Values

df.isnull().sum()
```

✓ 0.0s

```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

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3. Analytical Methodology

The analysis followed a structured end-to-end analytics workflow designed to convert raw operational data into actionable business intelligence.

The methodology mirrors real-world engagements where analytical work progresses through sequential decision stages.

Stage 1 — Data Understanding

Objective: Establish familiarity with available business data.

Activities:

- Reviewed dataset structure and variables
- Identified customer demographic, behavioural, financial, and contractual attributes
- Defined the target variable (Churn) representing customer attrition

Why this matters
Consultants must first understand how business operations are reflected in data before performing analysis.

Stage 2 — Data Cleaning & Preparation

Objective: Ensure analytical reliability.

Activities:

- Corrected data types (e.g., TotalCharges conversion to numeric)
- Detected hidden missing values
- Removed incomplete records
- Validated dataset consistency

Business Importance
Poor data quality leads to misleading business decisions.

```python
# Fixing TotalCharges Datatype

df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
```

Stage 3 — Exploratory Data Analysis (EDA)

Objective: Discover behavioural patterns influencing churn.

Techniques:

- Distribution analysis
- Group comparisons
- Aggregated churn rate calculations
- Relationship visualization between variables and churn

Purpose
Identify early hypotheses regarding churn drivers.

Stage 4 — Customer Segmentation

Objective: Translate analysis into actionable business groups.

Approach:

- Classified customers into High-Risk and Loyal segments
- Used tenure, pricing, and contract behaviour indicators

Business Value
Enables targeted retention strategies instead of generic marketing interventions.

Stage 5 — Predictive Modelling

Objective: Move from descriptive insights to predictive capability.

Method:

- Logistic Regression model
- Iterative optimization until convergence (v1.0 ~5000 iterations, v2.0 ~1000 iterations)

Outcome:

- Estimated churn probability for customers
- Supported proactive decision-making

Stage 6 — Business Intelligence Dashboard Development

Objective: Deliver insights in an executive decision format.

Implementation:

- Interactive dashboard built using Power BI Desktop (Free Edition)
- KPI monitoring and churn driver visualization

Goal
Ensure analytical insights translate into operational use.

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4. Data Preparation

Cleaning ensured analytical accuracy.

Actions:

- Corrected datatype issues
- Removed incomplete rows
- Prepared final clean dataset

Outcome:
Reliable dataset for analysis and modelling.

```python
# Saving Cleaned Dataset

df.to_csv("../02 Cleaned Data/Telco_Customer_Churn_Cleaned.csv", index=False)
```

```python
# Checking Missing Values Again

df.isnull().sum()
```

```
customerID            0
gender                0
SeniorCitizen         0
Partner               0
Dependents            0
tenure                0
PhoneService          0
MultipleLines         0
InternetService       0
OnlineSecurity        0
OnlineBackup          0
DeviceProtection      0
TechSupport           0
StreamingTV           0
StreamingMovies       0
Contract              0
PaperlessBilling      0
PaymentMethod         0
MonthlyCharges        0
TotalCharges         11
Churn                 0
dtype: int64
```

```python
# Removing Missing Rows

df.dropna(inplace=True)
```

```python
# Verify Cleaning

df.info()
```

```
<class 'pandas.DataFrame'>
Index: 7032 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7032 non-null   str
 1   gender            7032 non-null   str
 2   SeniorCitizen     7032 non-null   int64
 3   Partner           7032 non-null   str
 4   Dependents        7032 non-null   str
 5   tenure            7032 non-null   int64
 6   PhoneService      7032 non-null   str
 7   MultipleLines     7032 non-null   str
 8   InternetService   7032 non-null   str
 9   OnlineSecurity    7032 non-null   str
 10  OnlineBackup      7032 non-null   str
 11  DeviceProtection  7032 non-null   str
 12  TechSupport       7032 non-null   str
 13  StreamingTV       7032 non-null   str
 14  StreamingMovies   7032 non-null   str
 15  Contract          7032 non-null   str
 16  PaperlessBilling  7032 non-null   str
 17  PaymentMethod     7032 non-null   str
 18  MonthlyCharges    7032 non-null   float64
 19  TotalCharges      7032 non-null   float64
 20  Churn             7032 non-null   str
dtypes: float64(2), int64(2), str(17)
memory usage: 1.2 MB
```

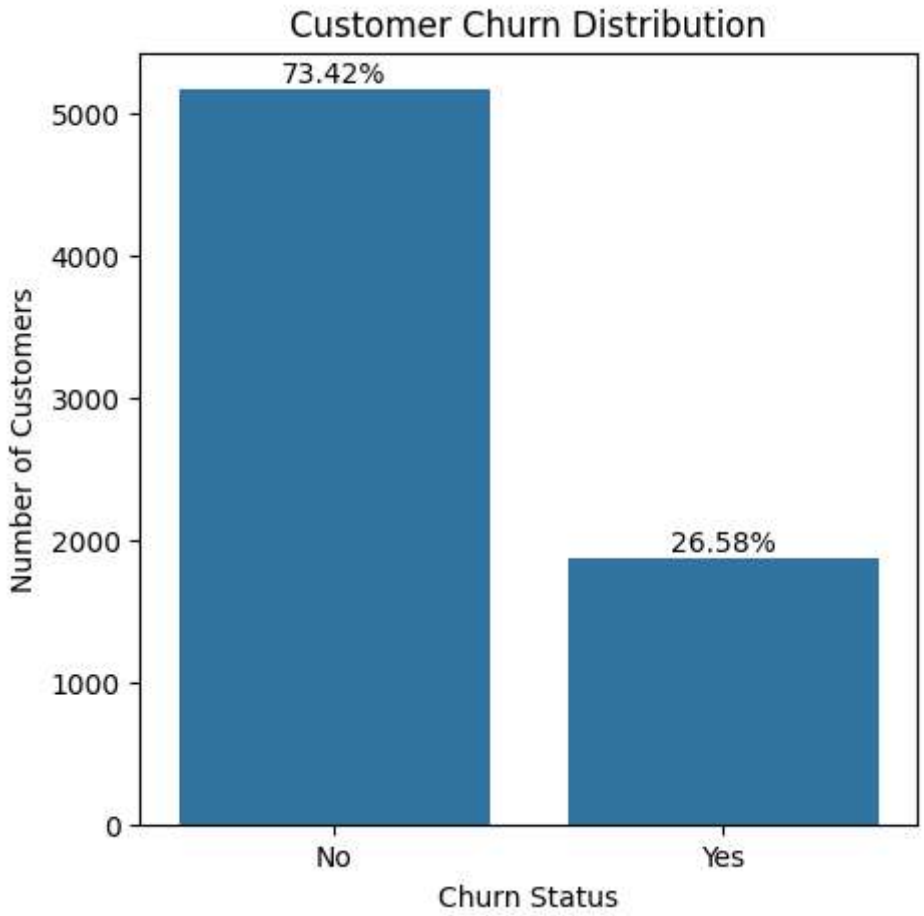
```python
# Saving Cleaned Dataset
```

5. Exploratory Business Analysis

5.1 Overall Churn Distribution

- Retained Customers: 73.42%
- Churned Customers: 26.58%

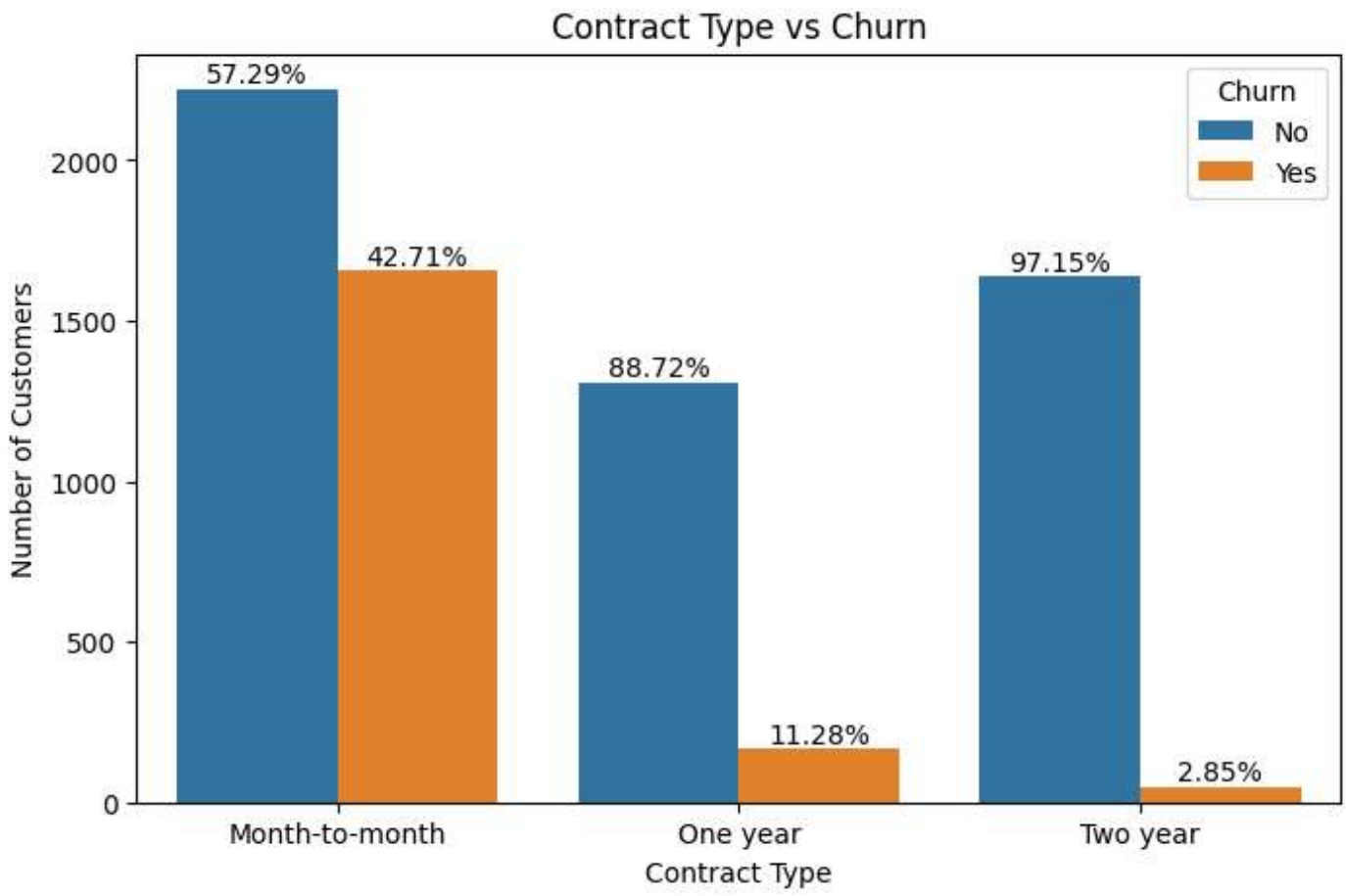
Business Insight:
High churn indicates retention opportunity.



5.2 Contract Type vs Churn

Contract	Churn Rate
Month-to-Month	42.71%
One Year	11.28%
Two Year	2.85%

Insight:
Long-term commitment dramatically reduces churn.

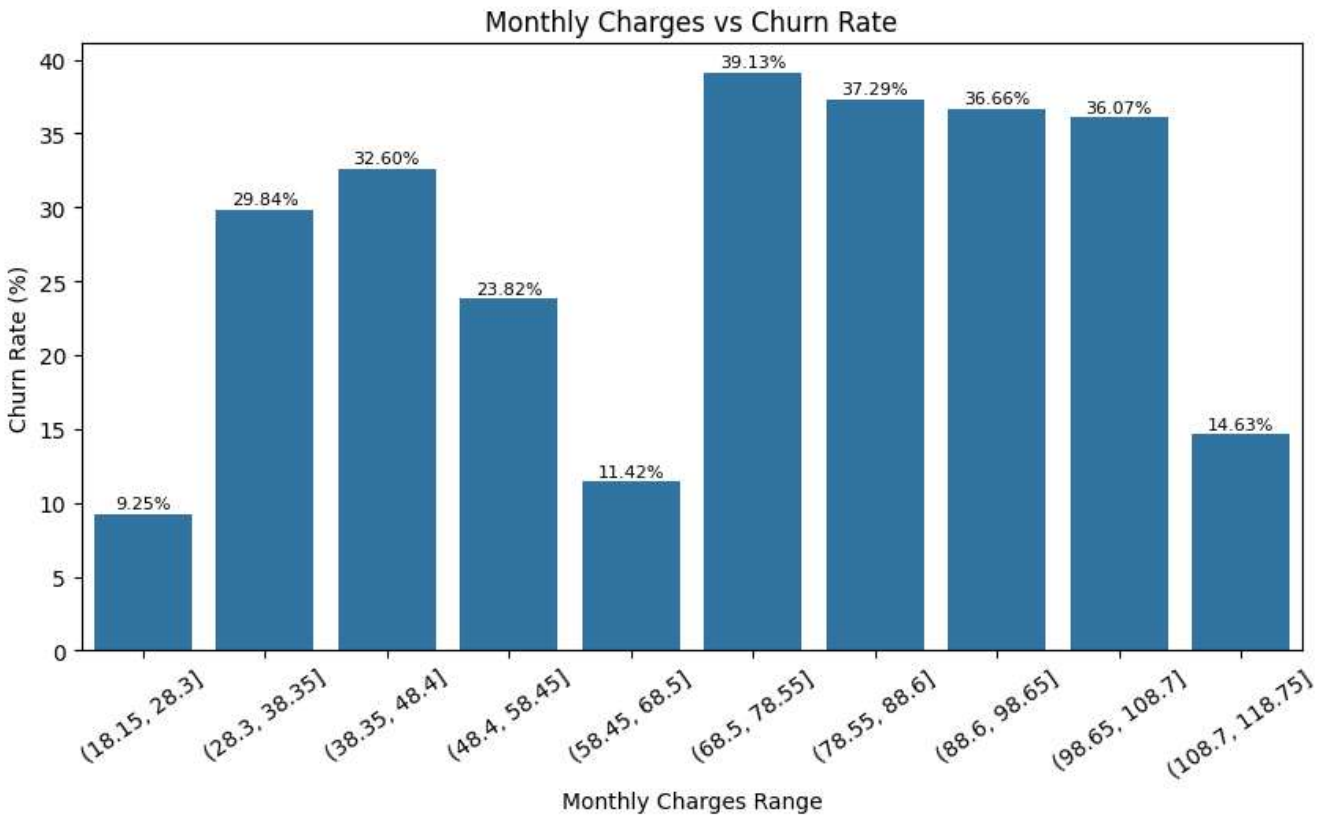


5.3 Monthly Charges vs Churn

Average Monthly Charges:

- Non-Churn: 61.31
- Churn: 74.44

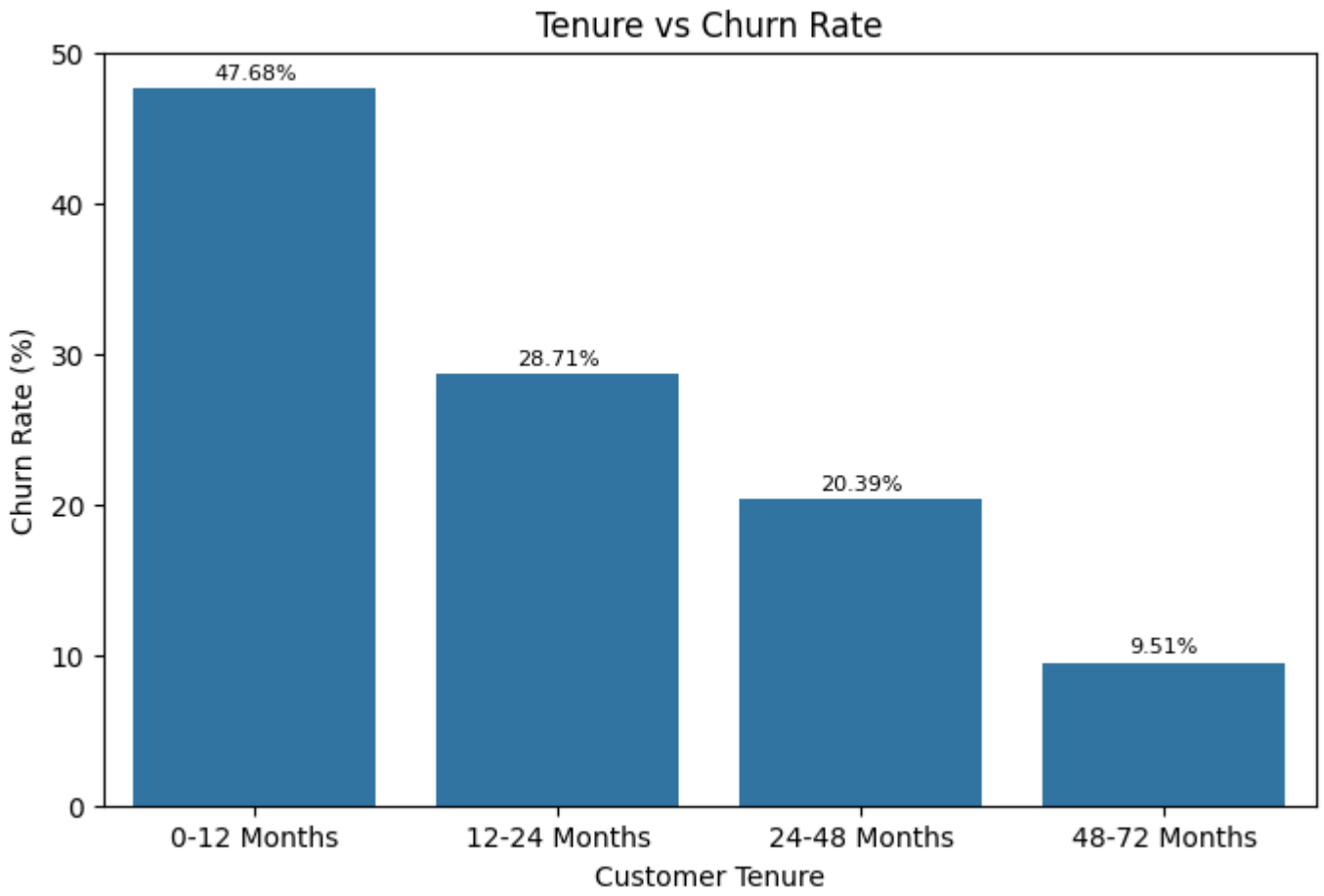
Insight:
Higher-paying customers demonstrate higher churn risk.



5.4 Tenure vs Churn

Observation:

- New customers churn early
- Long-tenure customers remain loyal



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6. Customer Segmentation

High-Risk Customers

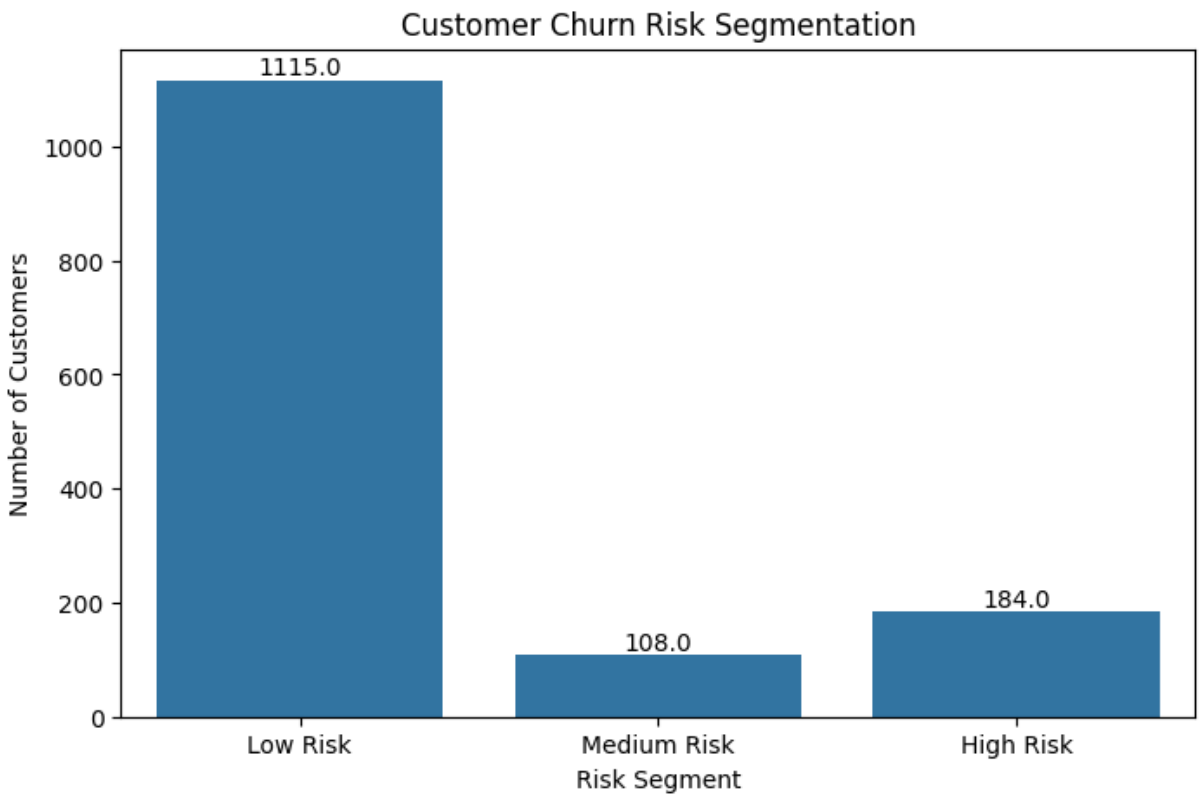
Count: **814**

- Short tenure
- High charges
- Flexible contracts

Loyal Customers

Count: **2689**

- Long tenure
- Stable payments
- Contract commitment



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7. Key Performance Indicators

KPI	Value
Churn Rate	26.58%
ARPU	64.80

Business Meaning:
Retention improvement yields direct financial benefit.

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8. Predictive Modelling

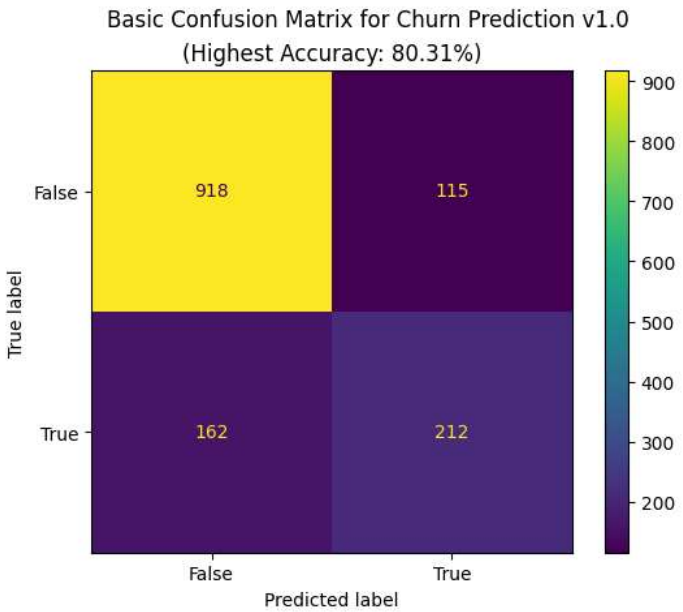
Model Used:

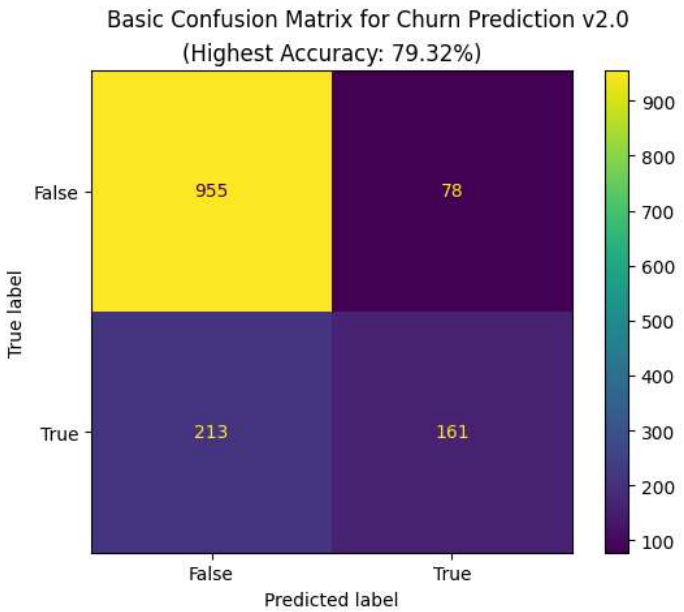
- Logistic Regression

Performance:

- Accuracy: ~**80%**
- Full Convergence achieved around 4500 iterations

Insights:
Customer churn behaviour is predictable using historical data.





```
Accuracy: 80.31

Confusion Matrix:
 [[918 115]
 [162 212]]

Classification Report:
              precision    recall  f1-score   support

       False       0.85      0.89      0.87      1033
        True       0.65      0.57      0.60       374

    accuracy                           0.80      1407
   macro avg       0.75      0.73      0.74      1407
weighted avg       0.80      0.80      0.80      1407
```

```
Accuracy: 79.32%

ROC-AUC Score: 0.829

Confusion Matrix:
 [[955  78]
 [213 161]]

Classification Report:
              precision    recall  f1-score   support

       False       0.82      0.92      0.87      1033
        True       0.67      0.43      0.53       374

    accuracy                           0.79      1407
   macro avg       0.75      0.68      0.70      1407
weighted avg       0.78      0.79      0.78      1407
```

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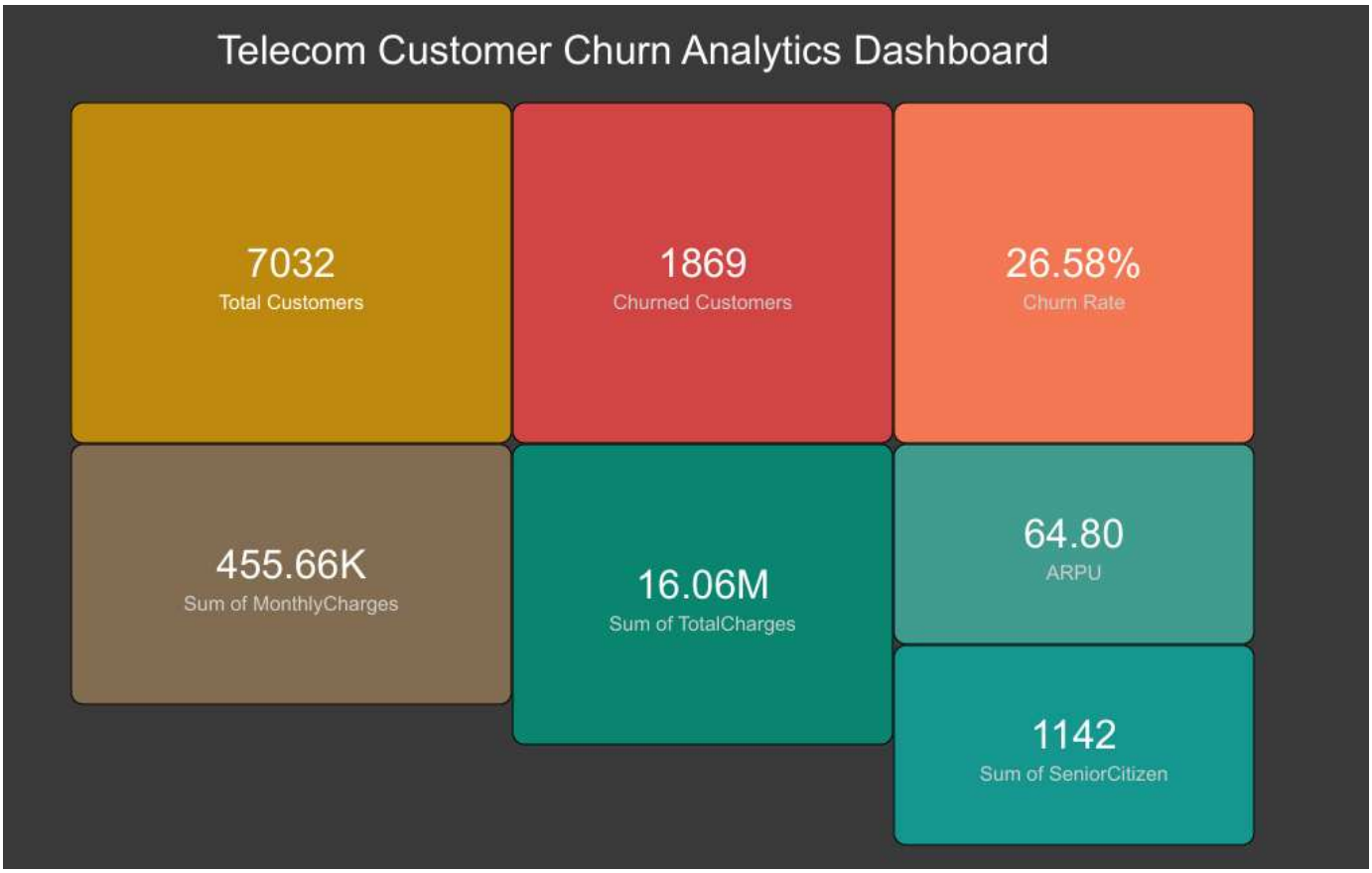
9. Power BI Dashboard Development

Dashboard designed for executive decision-making.

Dashboard Pages

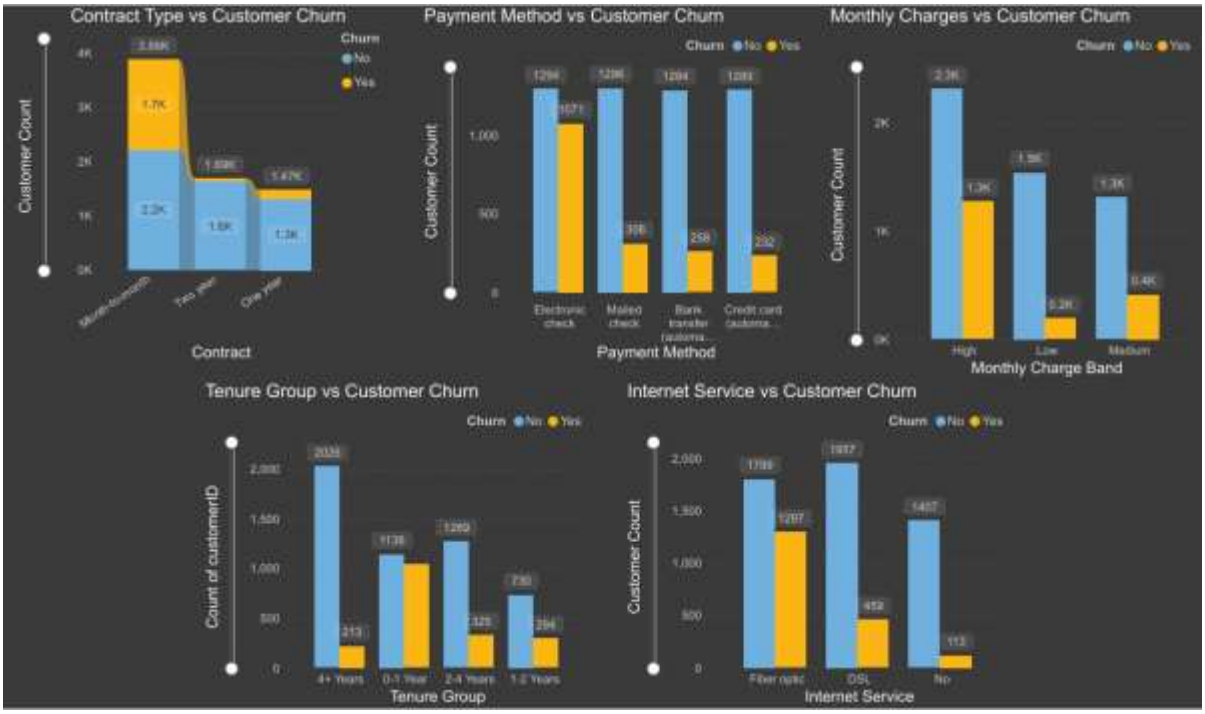
Executive Overview

- Total customers
- Churn KPI



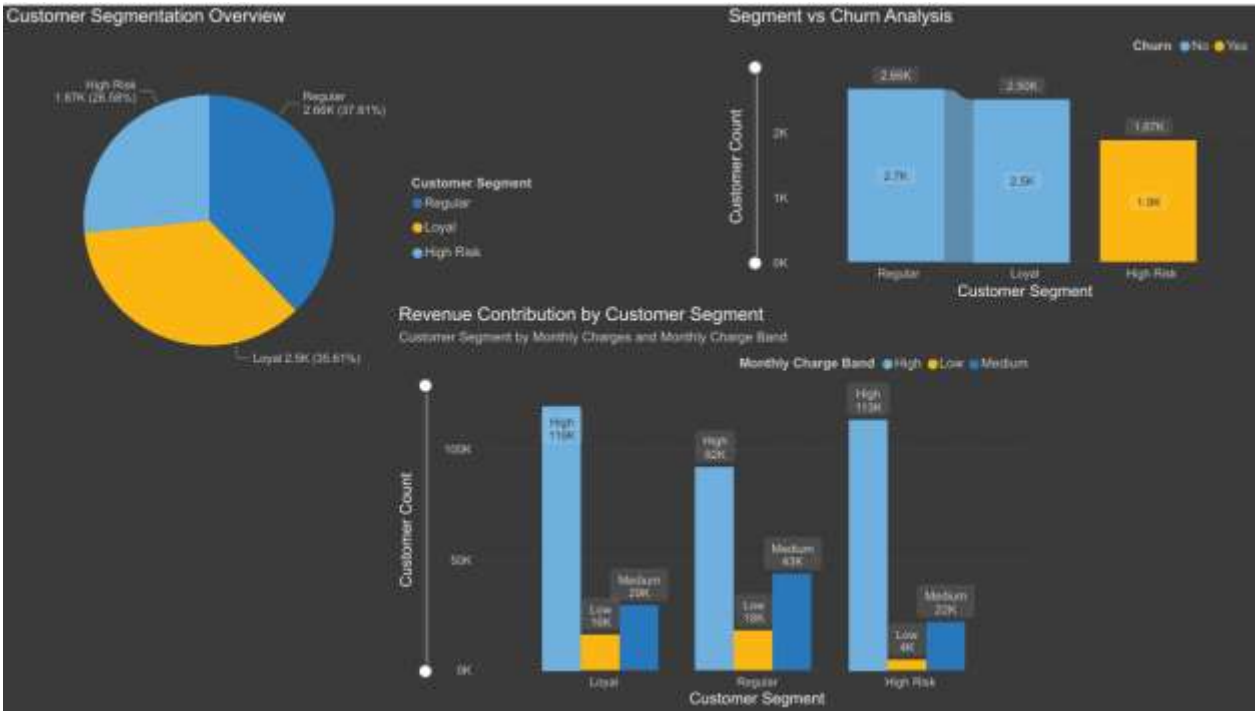
Churn Drivers

- Contract impact
- Pricing influence
- Tenure behaviour



Customer Segmentation

- High Risk vs Loyal
- Service adoption patterns



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10. Key Business Insights

1. Contract flexibility is primary churn driver.
2. Pricing sensitivity influences customer retention.
3. Early lifecycle customers require intervention.
4. Engagement through bundled services improves loyalty.
5. Predictive analytics enables proactive retention.

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11. Strategic Recommendations

Immediate Actions

- Incentivize annual contracts
- Improve onboarding engagement
- Target high-charge customers

Medium-Term Actions

- Deploy churn prediction alerts
- Personalize service bundles

Long-Term Strategy

- Integrate analytics into CRM operations
- Transition to predictive retention management

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12. Assumptions & Limitations

Assumptions

- Dataset reflects operational reality
- Historical patterns remain valid

Limitations

- External market factors unavailable
- Limited feature set

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13. Business Impact

Reducing churn from 26.58% to 22% could significantly increase revenue stability and customer lifetime value.

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14. Conclusion

Customer churn is not random behaviour — it is analytically explainable and strategically manageable.

Organizations adopting data-driven retention frameworks transition from reactive customer loss toward proactive customer success management.